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# Shahadat Hussain

**Date of birth:** 15/03/1990 | **Nationality:** Indian | **Gender:** Male | **Email address:** contact@shahadathussain.com | **Website:** www.shahadathussain.com | **LinkedIn:** shahadathussain | **Address:** Saadiyat Island, 129188, Abu Dhabi, United Arab Emirates (Work)

## ABOUT MYSELF

Driven Materials Scientist and R&D Engineer with over 7 years of specialized expertise in Industry 4.0 paradigms, focusing on advanced additive manufacturing, smart materials, and structural topology optimization. Documented research footprint across major transnational nodes (including New York University Abu Dhabi and Khalifa University), yielding 15+ peer-reviewed journal publications and over 450 citations (h-index: 10) in top-tier manufacturing literature.

**Core Technical Domain & Specializations:**

- **Additive Manufacturing:** Comprehensive mastery over process parameter optimization, metallurgical characterization, and heat treatment protocols for functional materials, carbon fiber-reinforced polymer composites, specifically Nitinol (NiTi) Shape Memory Alloys.
- **Topology-Optimized Metamaterials:**Deep expertise in the design, computational structural modeling, and mechanical characterization of Triply Periodic Minimal Surface (TPMS) architectures (Gyroid, Primitive, Diamond configurations) and multi-material Interpenetrating Phase Composites (IPCs).
- **Advanced Diagnostics:**Hands-on proficiency utilizing high-resolution materials characterization systems, including SEM, EBSD, XRD, DSC, and Instron mechanical testing systems to isolate and map complex process-structure-property relationships.

Leverages a verified track record as an international expert evaluator and referee for over 15 premier scientific journals (including *Additive Manufacturing*and *Materials & Design*). Positioned to deliver high-impact technical leadership, experimental precision, and advanced materials innovation within European industrial R&D sectors or elite institutional research consortiums.

## EDUCATION & TRAINING

11/08/2014 - 17/07/2019 - BHOPAL, INDIA

### **DOCTOR OF PHILOSOPHY IN ENGINEERING**- ACADEMY OF SCIENTIFIC AND INNOVATIVE RESEARCH

**Materials Science and Engineering**; Shape Memory Alloys (Cu-Al-Ni); Phase Transformations; Heat Treatment; Powder Metallurgy; Composite Science and Engineering; Characterization and Analytical Techniques (SEM, XRD, Optical Microscopy, DSC, Spectroscopy, Metallography);Mechanical Testing (Tensile, Compression, Fatigue and Vickers Hardness Testing), Hot Rolling, Lightweight Materials; Tribology; Nano Science; Technical Communications and Scientific Ethics

**Level in EQF:** 8

25/08/2008 - 08/06/2012 - BHOPAL, INDIA

### **BACHELOR OF ENGINEERING**- RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA

Mechanical Engineering | Industrial Training at Steel Authority of India Limited (SAIL)

**Level in EQF:** 6

## WORK EXPERIENCE

16/09/2024 - CURRENT - ABU DHABI, UNITED ARAB EMIRATES

### **RESEARCH ASSOCIATE**NEW YORK UNIVERSITY ABU DHABI

- Leading research on additive manufacturing of TPMS cubes and beams with composite filaments
- Design and characterization of TPMS metamaterials for structural applications
- Microstructural analysis (SEM, XRD, EDS) and mechanical testing (compression, bending)
- Developing additive manufacturing processes for interpenetrating phase composites (IPCs) and hybrid manufacturing

24/02/2020 - 30/06/2024 - ABU DHABI, UNITED ARAB EMIRATES

### **POSTDOCTORAL FELLOW**KHALIFA UNIVERSITY

- Leading research on additive manufacturing of TPMS NiTi shape memory alloys
- Design and characterization of TPMS metamaterials for biomedical and aerospace applications
- Microstructural analysis (SEM, XRD, EDS) and mechanical testing (tension, fatigue)
- Completed a 10-week Teaching Certificate Course at KU's Center for Teaching and Learning (2023)
- Completed Lab Safety and Responsible Conduct of Research (RCR) training

## SKILLS

ADDITIVE MANUFACTURING | laser powder bed fusion | Selective Laser Melting | Fused Deposition Method | Stereolithography | Fused Filament Fabrication | Mechanical Testing | Scanning Electron Microscopy | Materials Characterizations | MATLAB | Python | Data Science | Casting | Triply Periodic Minimal Surfaces | Architected Material | 3D Printing | 4D Printing | Shape Memory Alloys | NiTi | Advanced Materials | Materials Science | Interpenetrating Phase Composites | Hybrid Manufacturing | Green Metal 3D Printing

## LANGUAGE SKILLS

Mother tongue(s): **HINDI**

| | UNDERSTANDING | | SPEAKING | | WRITING |
|---|---|---|---|---|---|
| | Listening | Reading | Spoken production | Spoken interaction | |
| **ENGLISH** | C2 | C2 | C2 | C2 | C2 |

## CERTIFICATIONS

16/01/2026 Stanford University & DeepLearning.AI (Instructor: Andrew Ng)

**Machine Learning Specialization**

**Online - Core Competencies:** Supervised learning (linear/logistic regression, neural networks, decision trees), unsupervised learning (clustering, anomaly detection), and practical machine learning system design.

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**Link** https://www.coursera.org/account/accomplishments/certificate/P6EO92WGCSU2

23/08/2025 IBM

**IBM Data Science Professional Certificate**

**Online - Core Competencies:** Predictive data analysis, data visualization, Python-based scientific computing libraries (Pandas, NumPy, Scikit-learn), SQL database management, and exploratory data analysis.

**Link** https://www.coursera.org/account/accomplishments/specialization/certificate/4R8YJ5YWQ0NH

## CONFERENCES & SEMINARS

30/10/2022 - 03/11/2022 Greater Columbus Convention Center, Columbus, OH, USA

**ASME International Mechanical Engineering Congress and Exposition (IMECE 2022) Paper Title**: Inhomogeneous Microstructure due to Non-Uniform Solidification Rate in NiTi Triply Periodic Minimal Surface (TPMS) Structures Fabricated via Laser Powder Bed Fusion

**Role:** Presenter & Principal Lead Investigator

**Core Technical Contribution:**

- Investigated the localized phase transformation variations and solidification physics within complex 3D-printed shape memory alloy architectures.
- Demonstrated how non-uniform cooling gradients within complex minimal surface topologies alter structural properties during Laser Powder Bed Fusion (LPBF).
- Presented experimental evidence highlighting critical processing-window parameters needed to eliminate structural defects in smart metal lattices.

**Link** https://doi.org/10.1115/IMECE2022-95320

## HONOURS AND AWARDS

24/02/2020 Khalifa University, Abu Dhabi, UAE

**Postdoctoral Research Fellowship in Constitutive Modeling and Fatigue Prediction for 3D Printed Nitinol Smart Alloys Research** Awarded a highly competitive institutional fellowship to spearhead advanced research tracks in metal 3D printing. Developed and optimized Laser Powder Bed Fusion (LPBF) processing windows for Nitinol (NiTi) shape memory alloy architectures.

16/11/2016 Council of Scientific and Industrial Research (CSIR), New Delhi, India

- Awarded a highly competitive national fellowship based on a combination of performance in the Graduate Aptitude Test in Engineering (GATE) and rigorous technical evaluation by a national panel of experts.
- Secured full doctoral funding to execute advanced material science and metallurgy research tracks within the CSIR–Advanced Materials and Processes Research Institute (AMPRI).

10/08/2014 Council of Scientific and Industrial Research (CSIR), New Delhi, India

- Won an ultra-competitive national grant awarded to top-tier engineering graduates across India based on national GATE rank and technical merit.
- Secured full initial funding to launch core experimental setups, mechanical characterization pipelines, and metallurgical workflows within the CSIR–Advanced Materials and Processes Research Institute (AMPRI).

## PROJECTS

16/09/2024 - CURRENT

**Large-Scale 3D Printing of Composite Metamaterials Institutional Node:**

New York University Abu Dhabi (NYUAD) | 2024 – Present

**Core Objective:**

Design, additive manufacturing, and advanced characterization of structural lightweight metamaterials utilizing continuous/chopped carbon fiber-reinforced polymers (PLA-CF and PPA-CF).

**Technical Deployment:**

- **Design & Simulation:**Generated complex Triply Periodic Minimal Surface (TPMS) architectures, including Schwarz Primitive and Schoen Gyroid lattice topologies, validating structural responses via Finite Element Analysis (FEA) simulations.
- **Additive Manufacturing:**Operated industrial-tier Fused Filament Fabrication (FFF) setups (Raise3D Pro3 Plus HS, Stratasys F370, Bambu Lab X1) to print macro-scale structural elements, cubes, and beams.

- **Testing & Diagnostics:**Conducted extensive quasi-static compression and 4-point flexural testing to evaluate energy absorption (achieving up to 24–25 kJ/kg), stiffness limits, and dynamic vibration profiles (modal and frequency-response analysis).
- **Microstructural Quality Control:**Used Scanning Electron Microscopy (SEM) and optical microscopy to map fiber orientation, identify printing defects, and analyze interfacial interlayer bonding.

24/02/2020 - 30/06/2024

**Laser Powder Bed Fusion (LPBF) of Functional NiTi Smart Alloys  Institutional Node:**

Khalifa University, Abu Dhabi | 2020 – 2024

**Core Objective:**

Industrial-scale metal 3D printing of Nitinol (NiTi) shape memory alloy lattice structures and process-structure-property map optimization.

**Technical Deployment:**

- **Metal AM Operation:** Ran precision process-parameter loops on an industrial **EOS M400**LPBF system, systematically tuning laser power, scan speed, and laser incident angle to stabilize the dynamic melt pool.
- **Advanced Metallurgy & Diagnostics:**Evaluated local microstructural heterogeneity, grain orientation, and thermal phase transformation variations across complex geometric boundaries using FE-SEM, EBSD, X-ray Diffraction (XRD), and Differential Scanning Calorimetry (DSC).
- **Mechanical Damage Profiling:**Conducted strain-controlled cyclic compression and high-cycle fatigue testing (adhering to ASTM E606 / ISO 12106 standards) using Instron 8872/5969 universal testing machines to formulate structural constitutive models.

## PUBLICATIONS

**Do chopped fibers matter? Process-Structure-Property Relationships and Thermal Annealing Effects in Fiber-Reinforced TPMS Lattices** 2026

Triply periodic minimal surface (TPMS) lattices offer high structural efficiency for lightweight energy absorption, yet the effects of chopped fiber reinforcement and post-processing on their mechanical response remain poorly quantified. This study systematically investigates the mechanical behavior of additively manufactured gyroid TPMS lattices fabricated from five polymeric filaments —VeroWhite, Nylon 12, ABS, Carbon Fiber-reinforced PLA (PLA-CF) and polyphthalamide (PPA-CF)— across relative densities of 20–40%. Experimental results show that annealed fiber-reinforced lattices, particularly PPA-CF, consistently outperform unreinforced polymers in terms of peak stress, toughness, and energy absorption, with performance gains becoming more pronounced at higher relative densities. Post-print thermal annealing of PPA-CF increases its peak stress by 40% and toughness by up to 60%, while altering its post-yield deformation stability. Annealed PPA-CF gyroids achieve specific energy absorption values of up to 24–25 kJ/kg at densities of 200–430 kg/m$^3$, comparable to reported metallic TPMS structures, at substantially lower weight. Finite element analysis reveals elevated stress localization and increased plastic strains in fiber-reinforced lattices, consistent with the experimentally recorded post-elastic ductility trade-offs. These findings demonstrate that chopped fiber CF reinforcement, in combination with thermal post-processing, fundamentally modifies deformation behavior and enables high performance, lightweight TPMS architectures through coordinated control of relative density and processing.

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Shahadat Hussain, Haris Mehraj, Nikolaos Karathanasopoulos, Mohamed A. Moustafa Springer Nature Progress in Additive Manufacturing

**Phase Transformation Behavior of NiTi Triply Periodic Minimal Surface Lattices Fabricated by Laser PowderBed Fusion** 2024

Laser powder bed fusion (LPBF) is an appealing additive manufacturing technique suitable for producing intricate products with complex shapes, where traditional subtractive manufacturing methods may be impractical. When dealing with NiTi shape memory alloys, the fabrication process encounters additional challenges due to the material's high ductility and work hardening characteristics. These properties lead to elevated tool wear and poor workability. However, LPBF overcomes these challenges by eliminating the need for tooling and enabling direct fabrication of complex shapes based on predefined computer-aided designs. Although there has been a growing interest in the LPBF of NiTi in recent years, the fabrication of NiTi structures with architected designs has received limited attention. This study aims to bridge this research gap by investigating the phase transformation behavior of two specific types of triply periodic minimal surface (TPMS) architected NiTi lattices, namely primitive and gyroid, manufactured using LPBF. The study utilized a methodology involving the design of TPMS structures, additive manufacturing, and the characterization of printed samples through electron microscopy, x-ray diffraction, and thermal analysis to measure phase transformation temperatures. The research not only examines the impact of process parameters on the behavior of the fabricated samples but also establishes the influence of cellular geometry on the functional response of TPMS NiTi structures. It was noteworthy to observe that gyroid samples displayed a higher thermal hysteresis and lower reverse transformation enthalpy in comparison with primitive samples.

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Shahadat Hussain, Ali N Alagha, Wael Zaki Springer Nature Journal of Materials Engineering and Performance

### Microstructural and surface analysis of NiTi TPMS lattice sections fabricated by laser powder bed fusion

2023

Due to their high ductility and superior strength , the machining of NiTi shape memory alloys using conventional subtractive manufacturing technologies poses significant challenges. However, additive manufacturing (AM) offers a viable solution by circumventing the limitations of machinability through the elimination of tooling, thereby enabling the production of NiTi structures with previously unachievable intricacy. This study focuses on the fabrication of base layers of architected triply periodic minimal surface (TPMS) lattices using laser powder bed fusion (LPBF) and explores primitive and gyroid topologies. The investigation involves a comprehensive analysis and discussion of the influence of geometric properties and process parameters on the microstructural characteristics and the distribution of solid phases within the samples. Notably, the study reveals a substantial impact of process parameters and structural topology on the microstructural features . Additionally, notable observations are made concerning nickel evaporation, as well as the formation of oxide- and titanium-rich phases in relation to their distance from the base plate . Investigating intricate TPMS geometries in NiTi alloys in conjunction with varying laser process parameters represents a relatively new and unexplored area of research. These geometries may offer unique structural and functional properties that can potentially lead to innovative applications and advancements in various fields.

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Shahadat Hussain, Ali N Alagha, Gregory N. Haidemenopoulos, Wael Zaki Elsevier Journal of Manufacturing Processes 102:375-386, 2023.

### Imperfections Formation in Thin Layers of NiTi Triply Periodic Minimal Surface Lattices Fabricated UsingLaser Powder Bed Fusion

2022

In this paper, thin layers of NiTi shape memory alloy (SMA) triply periodic minimal surface lattices (TPMS) are fabricated using laser powder bed fusion (LPBF), considering different laser scanning strategies and relative densities. The obtained architected samples are studied using experimental methods to characterize their microstructural features, including the formation of cracks and balling imperfections. It is observed that balling is not only affected by the parameters of the fabrication process but also by structural characteristics, including the effective densities of the fabricated samples. In particular, it is reported here that higher densities of the TPMS geometries considered are generally associated with increased dimensions of balling imperfections. Moreover, scanning strategies at 45° angle with respect to the principal axes of the samples resulted in increased balling.

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Shahadat Hussain, Ali N Alagha, Wael Zaki MDPI Materials

### Inhomogeneous Microstructure due to NonUniform Solidification Rate in NiTi Triply Periodic Minimal Surface(TPMS) Structures Fabricated via Laser Powder Bed Fusion

2022

In recent times, interest in the fabrication of porous NiTi structures have grown significantly. Porous structures have remarkable potential to be used in the areas of tissue engineering, impact absorption, and fluid permeability. However, fabrication of NiTi structures poses challenges such as poor machinability, high work hardening, and inherent springback effects, which render them difficult to tackle through conventional manufacturing routes. Additive manufacturing (AM) can alleviate the aforementioned issues associated with NiTi shape memory alloys (SMAs). In addition, this technology can be employed for producing metallic scaffolds and porous structures of complex architectural details. Recently, a class of minimal surface topologies, known as triply periodic minimal surface (TPMS) structures has emerged as an attractive configuration for building architected constructs. Very little work can be found in the literature addressing the fabrication of NiTi TPMS structures and investigating their behaviors. The complex geometries of these structures may influence the dynamics of the melt pool in beam-based AM processes as well as the solidification rate within different regions of a product, thereby affecting the microstructures of fabricated parts. An inhomogeneity in microstructures of fabricated parts was observed, which motivated a detailed examination of these structures. The novelty of the present work lies in studying the influence of geometries of NiTi TPMS lattices along with laser process parameters.Write here the description

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Shahadat Hussain, Ali N Alagha, Wael Zaki ASME ASME International Mechanical Engineering Congress and Exposition (IMECE2022), Ohio, USA, 2022

## VOLUNTEERING

16/09/2024 - CURRENT NYU Abu Dhabi

- Providing hands-on training to students and researchers on operating Raise3D Pro3 Plus HS 3D printer for high-speed composite filament printing and Raise3D Forge1 3D printer for green steel (metal-polymer) printing.
- Training users in 3D model design, slicing software (ideaMaker), print parameter optimization, and post-processing techniques.
- Conducting sessions on basic maintenance, troubleshooting, and minor repair of the 3D printers to ensure continuous and reliable operation.
- Supporting the development of practical skills in advanced additive manufacturing technologies within the research group.

## NETWORKS AND MEMBERSHIPS

24/02/2020 - CURRENT Abu Dhabi

**International Peer-Review Refined Network Role:** Expert Independent Scientific Referee

**Vetting Bodies:***Additive Manufacturing*, *Materials & Design*, *Journal of Manufacturing Processes*, *Journal of Alloys and Compounds*, *Journal of Materials Engineering and Performance*(Elsevier, Springer, ASM International)

**Contribution:**Actively evaluating and refereeing advanced manuscripts in Laser Powder Bed Fusion (LPBF), metallurgy of shape memory alloys, and topology-optimized TPMS lattice configurations to enforce global scientific accuracy and innovation standards.

30/10/2022 - 03/11/2022 OH USA

**American Society of Mechanical Engineers (ASME) Role:** Published Presenter

**Core Engagement:**Permanently indexed within the ASME Digital Collection via technical conference tracks (such as the *ASME International Mechanical Engineering Congress and Exposition - IMECE*). Contributing to advanced manufacturing and structural mechanics dissemination.

11/08/2014 - CURRENT Digital

**Open-Science Global Research Identity Registries Network Affiliations:** ORCID Core Record (0000-0002-4964-9690) | Scopus Vetted Author Network (57310495100)

**Purpose:**Active node within interconnected global research graphs, maintaining fully auditable and machine-readable data streams of peer-reviewed publication impacts, metadata links, and institutional cross-referencing.

## OTHER

**Digital Research Identifiers & Global Archives**

Institutional Authentications & Research Registries

- **Verified Academic Affiliation Nodes:** * *Current Node:*Research Associate – New York University Abu Dhabi (NYUAD)
    - *Former Node:*Postdoctoral Fellow – Khalifa University, UAE

    - *Doctoral Node:*PhD Research Fellow – CSIR–Advanced Materials and Processes Research Institute (AMPRI), India
- **Persistent Digital Scholar Identifiers:*** *ORCID:*Connecting core publications, active metadata, and transnational research footprints.
    - *ISNI (International Standard Name Identifier):*Global authority identity layer utilized by international libraries and archives.
    - *Scopus Author ID & Web of Science ResearcherID:*Real-time tracking mechanisms validating 15+ peer-reviewed outputs, 450+ citations, and a peer-review network footprint.
- **Transnational Library & Open-Science Archives:**
    - *GND (Integrated Authority File):*Fully registered and mapped within the central authority architecture of the German National Library (DNB), serving Germany, Austria, and Switzerland.
    - *HAL Open-Access Repository:*Authenticated French national open-science network node (User ID: 1803496) supporting the decentralized, open dissemination of scientific literature.
    - *Global Bibliographic Registries:*Permanently indexed across WorldCat (OCLC), the British Library, NYU Libraries, and the NASA Science Explorer / Astrophysics Data System (ADS/SciX).